Review:

Evaluate the following conditions:
1 > 4 F
3 <= 9 T
1! == 0 T
“Cat” > ”cat” F
“Dog” == ”Zerba” F
True! == false T
-1>6 F
“elephant”>”Elephants” T

Evaluate the following conditions with logical operators:
1>2 && 3<8 F
0==0 || 7==9 T
!(3<11) F
“Cats”==”cats” && “dog”==”dog” F
!(“Cats”==”cats” && “dog”==”dog”) T
!(7<=7) && !(2==3) F
-8>=11 || “cat”==”catty” F

Exercise : Highest of Three

Write a program that takes in three numbers and returns the highest number.

```
let num1=+prompt("Enter the first number: 5");
let num2=+prompt("Enter the second number: 10");
let num3=+prompt("Enter the third number: 4");
if(num1>__num2_____ && num1>___num3_____)
{
console.log(num1 +" is the largest of the three.");
}
else if(num2>__num1_____ && num2>__mun3_____)
{
console.log(num2 +" is the largest of the three.");
}
else
{
```

```
console.log(_num3_________ +" is the largest of the three.");
}
```

Exercise : Sum Evens

Write a program that takes three integers and sums the value of those integers, but only if the value of the integers are even.

```
let num1=+prompt("Enter the first number: 2");
let num2=+prompt("Enter the second number:4 ");
let num3=+prompt("Enter the third number: 6 ");
let sum=0;
if(num1_ %__2==0 && num2_%__2==0 && num3__%_2==0)
{
   sum=num1+num2+num3;
   console.log(sum +" is the sum of these three  even  numbers.");
}
else
{
   console.log("All three numbers must be even to be added.");
}
```

Exercise : Nums in Order

Write a program that takes three numbers, and only returns true if all three numbers are input in increasing order.

```
let num1 = +prompt("Enter the first number: ");

let num2 = +prompt("Enter the second number: ");

let num3 = +prompt("Enter the third number: ");
```

```
if (num1 < num2 && num2 < num3) {
  console.log(true);
} else {
  console.log(false);
}
```

Exercise : Lucky 13

Write a program that takes in two values and outputs whichever number is closest to 13 without going over the value 13.

For example,inputs `9, 12` would output `12`, whereas `1, 14` would output `1`.

If both values are above 13, then output the value 0.

```
let a = +prompt("Enter the first number: ");
let b = +prompt("Enter the second number: ");
let result = 0;
// If both numbers are greater than 13, output 0
if (a > 13 && b > 13) {
  result = 0;
}
// If only 'a' is greater than 13, output 'b'
else if (a > 13) {
  result = b;
```

```
}
// If only 'b' is greater than 13, output 'a'
else if (b > 13) {
  result = a;
}
// If both numbers are less than or equal to 13,
// output the one that is closer to 13 (the larger one)
else {
  result = (a > b) ? a : b;
}
// Display the result
console.log(result);
```

Exercise : How Tired Am I?

Write a program that outputs `true` if the user is tired. A user is tired if the amount of sleep they get is less than 8 hours. If they've slept at least 5 hours, but have had coffee, the user is not tired.

```
let sleep = +prompt("How many hours did you sleep? ");
let coffee = prompt("Did you have coffee? (yes/no)");
if (sleep < 8 && !(sleep >= 5 && coffee.toLowerCase() === "yes")) {
  console.log(true); // tired
} else {
  console.log(false); // not tired
```

```
}
```

Exercise : Your Grade?

- Write a program to input marks of five subjects Physics, Chemistry, Biology, Mathematics and Computer, calculate percentage and grade according to given conditions:
  If percentage >= 90% : Grade A
  If percentage >= 80% : Grade B
  If percentage >= 70% : Grade C
  If percentage >= 60% : Grade D
  If percentage >= 40% : Grade E
  If percentage < 40% : Grade F

Example

Input

```
Input marks of five subjects: 95 95 97 98 90
```

Output

```
Percentage = 95.00 Grade A
```

```
let phy = +prompt("Enter marks for Physics: ");
let chem = +prompt("Enter marks for Chemistry: ");
let bio = +prompt("Enter marks for Biology: ");
let math = +prompt("Enter marks for Mathematics: ");
let comp = +prompt("Enter marks for Computer: ");

let percentage = (phy + chem + bio + math + comp) / 5;

let grade = "";
if (percentage >= 90) grade = "A";
else if (percentage >= 80) grade = "B";
else if (percentage >= 70) grade = "C";
else if (percentage >= 60) grade = "D";
else if (percentage >= 40) grade = "E";
else grade = "F";

console.log("Percentage = " + percentage.toFixed(2) + "%, Grade " + grade);
```

Exercise : Bills?

- Write a program to input electricity unit charge and calculate the total electricity bill according to the given condition:
  For first 50 units Rs. 0.50/unit
  For next 100 units Rs. 0.75/unit
  For next 100 units Rs. 1.20/unit
  For unit above 250 Rs. 1.50/unit
  An additional surcharge of 20% is added to the bill.

```
let units = +prompt("Enter electricity units: ");
let amount = 0;

if (units <= 50) {
  amount = units * 0.50;
} else if (units <= 150) {
  amount = 50 * 0.50 + (units - 50) * 0.75;
} else if (units <= 250) {
  amount = 50 * 0.50 + 100 * 0.75 + (units - 150) * 1.20;
} else {
  amount = 50 * 0.50 + 100 * 0.75 + 100 * 1.20 + (units - 250) * 1.50;
}

let total = amount + amount * 0.20; // 20% surcharge
console.log("Total bill = Rs. " + total.toFixed(2));
```

Exercise : The Cartesian Coordinate

Write a program to accept a coordinate point in a XY coordinate system and determine in which quadrant the coordinate point lies.
Test Data : 7 9
Expected Output :
The coordinate point (7,9) lies in the First quadrant.

```
let x = +prompt("Enter x coordinate: ");
let y = +prompt("Enter y coordinate: ");

if (x > 0 && y > 0) {
  console.log("The point (" + x + "," + y + ") lies in the First quadrant.");
} else if (x < 0 && y > 0) {
  console.log("The point (" + x + "," + y + ") lies in the Second quadrant.");
} else if (x < 0 && y < 0) {
  console.log("The point (" + x + "," + y + ") lies in the Third quadrant.");
} else if (x > 0 && y < 0) {
  console.log("The point (" + x + "," + y + ") lies in the Fourth quadrant.");
} else {
  console.log("The point (" + x + "," + y + ") lies on an axis.");
}
```